

 MIT Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)		ACTIVITY / ACTIVITIES	
Alandi (D), Pune - 412105		ACADEMIC YEAR	2026 - 2027
DEPARTMENT OF E&TC ENGINEERING		SEMESTER	VII
		CLASS	BTech
		DIVISION	C
COURSE CODE	2307476T	ACTIVITY NO.	
COURSE	PROJECT MANAGEMENT	DATE	
ACTIVITY DETAILS			
Name of Activity	"GitHub-Based QA Documentation and Problem Solving based on Electronics/Cross Domain Projects"		
Activity Details	Create a basic GitHub repository for an embedded system code (e.g., LED Blinking using Arduino), log QA issues using GitHub Issues, and collaborate by commenting or resolving them. Prepare a small report and upload on Moodle A. GitHub Repository Link: B. Brief Description of Code C. QA Issues Logged D. Collaboration Summary E. Learning Outcome F. Prepare Report and upload on Moodle G. Project Planning and Tracking		
Type of Activity	Individual	Mode of Assessment	Online
Course Topic Covered	Implementation of Collaborative tools	Marks	15
Course Outcome	CO2	RBT Level	Level 3
No of Present Students			
Learning Styles Addressed	L3		
Pre-Reflection of Activity	Consider how collaborative platforms like GitHub can enhance transparency, traceability, and teamwork in solving real-world electronics and interdisciplinary project challenges.		
Post-Reflection of Activity	Reflect on how GitHub helped streamline documentation and solve project issues collaboratively.		
No of Weaker Students			
No of Bright Students			
No of Average Students			
Geo tag Photo			

(Remark: Course Instructor to add assessment rubrics for each activity)

Criteria	CO Addressed	Exceptional Quality	Best Quality	Average Quality	Low Quality	Lowest Quality
		(5 Points)	(4 Points)	(3 Points)	(2 Point)	(1 Point)
Analysis of QA Documentation & GitHub Usage	CO2	Thoroughly analyzes QA documentation structure; correctly uses GitHub features (Issues, Wiki, Pull	Analyzes QA documentation; properly uses 2-3 GitHub features; documents 3-4	Attempts QA analysis; uses GitHub but inconsistently; documents 2-3	Limited QA analysis; minimal GitHub feature usage; vague documentation structure	No QA documentation analysis or GitHub usage shown
Problem Identification & Root Cause Analysis	CO2	Identifies 4+ problems in electronics/cross-domain project; performs thorough root cause analysis using 5-Why or fishbone; traces issues	Identifies 3-4 problems; performs root cause analysis; most issues traced in GitHub; clear severity	Identifies 2-3 problems; attempts root cause analysis; partial GitHub tracing; basic severity assessment	Identifies 1-2 problems; minimal root cause analysis; poor GitHub documentation	No problem identification or analysis demonstrated
Solution Implementation & Documentation	CO2	Proposes well-documented solutions for each problem; uses GitHub commits/branches effectively; creates	Proposes solutions with documentation; uses GitHub features (commits/branch	Attempts solutions; basic GitHub documentation; partial resolution records; limited testing evidence	Vague solutions; poor GitHub documentation; minimal resolution tracking	No solutions or implementation documented

Dr Ashish Mulajkar

COURSE TEACHER
Name and Sign with date

ACTIVITY DETAILS			
Name of Activity	"COCOMO vs. BoM/BoQ: Analysis of Real Time Electronics Project Planning"		
Activity Details	1: Select a Mini Embedded Project 2: Estimate Software Effort using COCOMO 3: Prepare Digital BoM/BoQ 4: Analyze and Compare your work 5: Submit Your Work using Glibli Images on A3 Size paper, this is a group activity in 3 students		
Type of Activity	Collaborative	Mode of Assessment	Online
Course Topic Covered	CH5	Marks	15
Course Outcome	CO4	RBT Level	Level 4
No of Present Students			
Learning Styles Addressed	L4		
Pre-Reflection of Activity	Review the COCOMO (Constructive Cost Model) basics: cost estimation, effort calculation, and productivity factors.		
Post-Reflection of Activity	Students must apply both models (COCOMO for software, BoM/BoQ for hardware) and then reflect on which gave more actionable insights		
No of Weaker Students			
No of Bright Students			
No of Average Students			
Geo tag Photo			

(Remark: Course Instructor to add assessment rubrics for each activity)

Criteria	CO Addressed	Exceptional Quality	Best Quality	Average Quality	Low Quality	Lowest Quality
		(5 Points)	(4 Points)	(3 Points)	(2 Point)	(1 Point)
Understanding & Comparison of COCOMO vs BoM/BoQ	CO4	Excellent explains COCOMO fundamentals (Basic, Intermediate, Detailed modes); clearly contrasts with BoM/BoQ; identifies 4+ key differences;	Well explains COCOMO model; compares with BoM/BoQ; identifies 3-4 key differences; gives relevant examples	Explains COCOMO concepts; attempts comparison; identifies 2-3 differences; examples are vague	Limited understanding of models; minimal comparison; vague differences identified	No clear understanding or comparison demonstrated
Application to Real-Time Electronics Projects	CO4	Expertly applies both models to real electronics scenarios; demonstrates when to use each; calculates cost/effort estimates accurately; shows practical constraints consideration	Applies models to electronics examples; explains when each is useful; mostly accurate calculations; considers some constraints	Attempts application to electronics projects; partial understanding of when to use models; calculation errors present	Poor application; minimal calculation; limited practical connection to real-time projects	No application or examples provided

Synthesis & Integrated Approach Recommendation	CO4	Proposes innovative hybrid cost model combining COCOMO + BoM/BoQ strengths; justifies with strong reasoning; addresses real-time project challenges; provides implementation roadmap	Suggests practical integration approach; combines elements from both models; good justification; addresses most project needs	Mentions possibility of integration; weak synthesis; limited justification for approach; incomplete roadmap	Minimal synthesis effort; poor justification; unclear recommendations	No synthesis or recommendations provided
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